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Original research

The effect of fresh and used ankle taping on lower limb biomechanics in sports specific movements

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ABSTRACT

Objectives: We aimed to investigate the effects of ankle taping on lower extremity biomechanics related to injury development and how these effects change after sports-specific use.
Design: Randomized, repeated measures design with three conditions: Barefoot, tape applied fresh, and tape after sports-specific use (between-subject factor: sex).
Methods: Twenty-five healthy participants (ten female) performed sports-specific movements, including running, drop jumping, and 180° change of direction, under the three conditions. Kinetic and kinematic data were collected using 3D motion capturing and force platforms.
Results: Tape applied fresh and tape after sports-specific use significantly reduced peak ankle inversion. Biomechanical risk factors for anterior cruciate ligament or running overuse injuries were either unchanged or decreased with tape applied fresh, except for the peak loading rate of the resultant ground reaction force, which increased between 4% and 18% between movement types. After 15 minutes of sports-specific use of the tape, the alterations induced by tape applied fresh remained for some biomechanical risk factors while they became closer to barefoot again for others, indicating a differential response to prolonged use of taping for different biomechanical variables.
Conclusions: Ankle taping protects the ankle joint by reducing biomechanical risk factors associated with ankle sprains, and most biomechanical risk factors for anterior cruciate ligament or running overuse injuries are not increased. Further research is needed to explore the duration of protective effects, variations across sports, and its impact on patients with chronic ankle instability, contributing to a more comprehensive understanding of ankle taping's influence on lower extremity biomechanics.
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Practical implications

- Newly applied ankle taping provides significant protection against ankle sprains by effectively reducing peak ankle inversion angles during dynamic sports-specific movements.
- Ankle taping does not significantly increase most biomechanical risk factors for ACL or running overuse injuries, making it a viable preventive measure for ankle sprains without substantially raising the risk of most other injuries.
- Despite its protective benefits, ankle taping increases peak ground reaction force loading rates, indicating a potential elevated risk for overuse injuries such as plantar fasciitis.
- The restrictive effects of ankle taping on peak ankle inversion angles persist to a large extent even after 15 min of sports-specific use, and future studies need to explore the effects of more prolonged usage.

1. Introduction

Ankle injuries are frequently encountered by athletes, which may cause a break in sports participation.[1] Ankle injuries can cause various problems ranging from simple sprains to chronic ankle instabilities and cartilage loss.[2] The most common presentation is the lateral ankle sprain. Recurrent ankle sprains can keep athletes off the field for long periods, sometimes even forcing them to stop playing professionally.[3]

Ankle taping is a commonly used method to prevent first and recurrent injuries.[4] Ankle sports taping reduces the risk of ankle injuries during, e.g., landing, sudden acceleration, and change of direction by reducing peak ankle excursions and, thereby, the strain experienced by ankle ligaments.[5,6] However, the taping-induced kinematic changes

Abbreviations: ACL, anterior cruciate ligament; BF, barefoot; COD, 180° change of direction; DJ, drop jumping; ES, effect size; GRF, ground reaction force; ROM, range of motion; RUN, running; TF, tape applied fresh; TU, tape after sports-specific use.

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might alter the ground reaction force (GRF) application (e.g., by increasing loading rates) or lower extremity joint kinematics and kinetics, potentially increasing the risk of lower extremity injuries.[7] The increased injury risk due to taping might relate to traumatic injuries (e.g., anterior cruciate ligament [ACL] injury) during high-risk movements (e.g., dynamic landing or changing direction) or to overuse injuries during frequently occurring movements like running.

Due to its immediate effects on lateral ankle ligament strain, peak ankle inversion is considered a biomechanical risk factor for lateral ankle strain injuries.[8] While restricting peak ankle inversion through taping can improve protection, reduced joint excursion in the frontal and sagittal planes (i.e., reduced ankle dorsiflexion) might increase GRF loading rates associated with several overuse-type injuries.[7] Consequently, these parameters should be included in assessing the effects of ankle taping during sports-specific movements, like change of direction or drop-jumping tasks.

Several parameters must be considered when considering changes in ACL injury risk when taping ankle joints. Increased knee abduction angles and internal knee adduction moments have been related to ACL injury risk.[9] However, proximal changes in the kinematic chain can affect these knee biomechanics, e.g., trunk and hip joint mechanics in the frontal plane of movement.[10] Therefore, when considering the effects of taping on risk factors for ACL injury, peak hip abduction moments and adduction angles should be considered.

Since running forms a significant part of any team sport, it should be interesting to see how a taping intervention affects biomechanical risk factors for common overuse injuries of the lower extremities (e.g., plantar fasciitis, iliotibial band syndrome, or medial tibial stress syndrome) during running. A recent systematic review summarized the most likely biomechanical risk factors for the most common overuse injuries in distance running,[7] which serve as the basis for assessing the effects of taping on running-related overuse injury risk.

Although the effects of taping on joint kinematics are relatively well reported, it is not yet clear how these changes in joint angles affect lower limb kinetics, particularly for more proximal joints, i.e., the knee and hip. A systematic review by Megalaa and colleagues has identified insufficient data to draw conclusions on this issue.[1] Furthermore, different taping methodologies add variability to the findings in the literature.[5,11] It is also known that the ankle excursion-restrictive properties of taping are reduced after 15 min of sports-specific exercises.[12] Yet, in detail, how these changes affect lower limb kinetics (in particular knee and hip biomechanics) remains unclear.[13]

When protecting the ankle, tape interventions interact with the stabilizing characteristics of athletic footwear. A previous study has shown that the restrictive effects of taping vary between shod and barefoot conditions.[14] However, athletic footwear varies substantially between sports and manufacturers, which results in large variability in potential footwear-by-tape interactions. Furthermore, there is a lack of correlation between changes in joint kinematics and data collected from shoe-mounted markers.[15] For example, Reinschmidt and colleagues found statistically significant differences between in-shoe and on-shoe kinematics, particularly for the frontal plane, which can lead to miscalculations of the biomechanical measurements.[16] Therefore, when analyzed within athletic footwear, a precise understanding of the isolated effects of taping interventions on ankle biomechanics might be hard to reach.

Consequently, this study aims to determine (1) whether there are changes in the kinetics and kinematics related to traumatic or overuse lower extremity joints during sports-specific tasks with and without taping during barefoot movements and (2) whether there will be any change in the taping-induced kinetic and kinematic alterations of the lower extremities obtained after using the tape for a specific time. We hypothesize that (I) taping reduces peak joint excursions in the sagittal and frontal plane in the ankle joint; (II) taping alters ankle, knee, and hip joint angles and moments associated with ACL or running overuse injury risk for these planes; and (III) that these effects will be reduced after using the tape during sports-specific-movements for a longer time (15 min).

2. Methods

Twenty-five healthy participants (ten women and fifteen men) participated in the study (age: 24.8 ± 2.5 years, mass: 73.0 ± 10.1 kg, height: 1.76 ± 0.99 m). At the time of the study, participants were free of injuries and pain and had no lower extremity injury in the last six months and no ankle injury in the last 12 months. Before participating in the study, all participants were informed about the procedures and gave written informed consent. The study procedures were approved by the ethics committee of Offenburg University of Applied Sciences, Offenburg, Germany (Approval Number: 04/23).

All participants performed warm-up and stretching exercises for approximately 10 min. The testing conditions were defined as barefoot (BF), tape applied fresh (TF), and tape after sports-specific use (TU). Randomization was applied between the BF and TF conditions to avoid fatigue bias. TU was always performed after the TF condition. For the tape conditions, a standard method of ankle taping was applied.[11,17] First, the area was wrapped with pre-wrap material (M Wrap, Mueller®, USA). Then, the start and end points were fixed with tape (EuroTape Platinum Grade 3.8 cm ∗ 11.4 m, Mueller®, USA) by applying standard stirrup strips, horseshoe strips, heel locks, eight-figure locks, and circumferential strip locks. The same sports physician with ten years of clinical experience in sports medicine ensured that the tape application was consistent for all participants. To mimic the effect of sports-specific movements on ankle taping, participants performed standardized running, sudden acceleration, change of direction, and jumping movements with their own sports shoes after the TF data collection phase. Each movement was performed for 3 min. Before starting a new movement, the participants engaged in 1 min of high-speed walking. The sports-specific use phase of the tape lasted 15 min for every participant. Subsequently, we collected data on the TU condition without athletic footwear.

The movements for each condition were performed on a tartan floor (CONICA AG, Zürich, Switzerland) without shoes. Drop jumping (DJ; 25 cm height), running (RUN; 4.5 ± 0.2 m/s), and 180° change of direction (COD; approach speed 3.0 ± 0.2 m/s) were performed as sports-specific movements for every condition in the same order. For RUN and COD, participants had to match the desired running speed and strike the force platform with the right foot without targeting. A digital wireless double light barrier system measured average RUN and COD approach speeds (Witty®, Microgate, Italy). For DJ, trials with no apparent anteroposterior or mediolateral displacement of the body between take-off and landing were considered valid. Data were collected for three valid trials per movement and condition.

Kinematic data were collected with 22 marker-based tracking cameras (200 Hz, Miqus, Qualisys AB, Gothenburg, Sweden). Kinetic data were collected through two 0.6 × 0.9 m force platforms embedded in the floor (1000 Hz, BMS60900, AMTI, MA; USA). Spherical retroreflective markers (diameter: 14 mm) were used to track the motion of the right lower extremity. Markers were placed on the pelvis, thigh, shank, rearfoot, and midfoot of the right lower extremity of each participant. Anatomical markers were placed on both posterior and anterior superior iliac spines and the medial and lateral femoral condyle, medial and lateral malleolus, medial, lateral, and posterior calcaneus, first and fifth metatarsophalangeal joints, and on top of the big toe on the right lower extremity. Additional tracking markers (four markers per segment) were attached to the skin to the mid-sections of the thigh and shank on the lateral side of the right lower extremity.[18] Marker coordinates of each segment were mathematically optimized to improve their compliance with rigid body assumptions.[19]

The marker and ground reaction force data were filtered using a low-pass 4th-order digital Butterworth filter with a cut-off frequency

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Table 1
Mean ± 1 standard deviation of analyzed parameters for the change of direction movement.

Parameter	Barefoot (BF)	Tape fresh (TF)	Tape used (TU)	p-Value Tape	p-Value Sex	p-Value Interaction
Peak ankle inversion angle (°)	15,4 ± 3,9$^{TF***, TU***}$	11,3 ± 4,1$^{BF***, TU*}$	12,6 ± 3,5$^{BF***, TF*}$	**<0.001**	**0,049**	0,484
Peak internal ankle inversion moment (Nm/kg)	0,38 ± 0,30^{TU*}	0,30 ± 0,17	0,27 ± 0,18^{BF*}	**0,046**	0,558	0,773
Peak ankle dorsiflexion angle (°)	−20,4 ± 6,9^{TF**}	−16,3 ± 5,5^{BF**}	−18,4 ± 5,7	**0,021**	0,163	0,352
Peak knee abduction angle (°)	−4,9 ± 7,1	−5,0 ± 6,8	−6,3 ± 6,6	0,101	**0,010**	0,418
Peak internal knee adduction moment (Nm/kg)	0,54 ± 0,27	0,55 ± 0,36	0,53 ± 0,27	0,746	0,627	0,113
Peak hip adduction angle (°)	−13,2 ± 7,1	−13,6 ± 8,3	−13,1 ± 7,0	0,686	0,600	0,088
Peak internal hip abduction moment (Nm/kg)	−0,82 ± 0,39	−0,79 ± 0,39	−0,75 ± 0,45	0,320	0,899	0,141
Peak resultant GRF (N/kg)	21,3 ± 3,4	22,4 ± 4,6	22,0 ± 4,1	0,059	0,078	0,428
Peak loading rate of res. GRF (N/kg/s)	544,5 ± 225,6$^{TF**, TU***}$	643,5 ± 214,8^{BF**}	675,9 ± 245,8^{BF***}	**<0.001**	0,318	0,318

Note: Bold parameter names indicate a significant ANOVA effect ($p < 0.05$). *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$ revealing significant Bonferroni-corrected post-hoc tests. p-Value adjusted for comparing a family of 3.

of 20 Hz.[20,21] The data from the filtered markers were then implemented into a six-degrees-of-freedom, five-segment rigid-body model of the lower extremities (including rearfoot, forefoot, shank, thigh, and pelvis segments). We characterized the motion of the ankle joint by analyzing the relative movement between the rearfoot and the shank segments. Joint angles were determined using Cardan angles derived from the rotational matrices that connect the distal to the proximal segments, following a sequence of flexion–extension, inversion–eversion/adduction–abduction, and internal–external rotation. Joint moments were expressed as internal moments within the proximal segments' anatomical reference frames. Further details of the model can be found in previous publications.[18,22]

For each trial, we extracted parameters related to either traumatic (ankle sprain or ACL injury during COD and DJ) or overuse injury risk (during RUN[7]; see Tables 1, 2 and 3 for a complete list of analyzed parameters). We averaged these parameters across the three trials obtained for each condition and motion for further statistical analysis. To identify differences between conditions, we used repeated measures ANOVAs with sex as a between-subject factor. In the presence of a significant main or interaction effect, we performed Bonferroni adjusted post-hoc dependent sample *t*-tests for a more detailed analysis of the differences between conditions. Cohen's d values were calculated as a measure of effect size (ES) in case of significant main or interaction effects.[23] The alpha level was set to 5 % for all statistical tests.

Table 2
Mean ± 1 standard deviation of analyzed parameters for the drop jump movement.

Parameter	Barefoot (BF)	Tape fresh (TF)	Tape used (TU)	p-Value Tape	p-Value Sex	p-Value Interaction
Peak ankle inversion angle (°)	12,3 ± 3,8$^{TF***, TU*}$	9,1 ± 3,1^{BF***}	10,4 ± 3,5^{BF*}	**<0.001**	0,302	0,537
Peak internal ankle inversion moment (Nm/kg)	0,85 ± 0,42$^{TF**, TU**}$	0,71 ± 0,34^{BF**}	0,70 ± 0,31^{BF**}	**0,002**	0,065	0,264
Peak ankle dorsiflexion angle (°)	−20,2 ± 7,3	−17,9 ± 8,9	−19,9 ± 7,4	0,252	0,840	0,549
Peak knee abduction angle (°)	−5,7 ± 6,1^{TU**}	−6,1 ± 6,8	−7,2 ± 7,3^{BF**}	**0,011**	**0,001**	**0,014**
Peak internal knee adduction moment (Nm/kg)	0,56 ± 0,26	0,56 ± 0,27	0,58 ± 0,29	0,387	**0,020**	0,165
Peak hip adduction angle (°)	−3,56 ± 4,61	−3,31 ± 4,93	−3,23 ± 4,63	0,921	0,389	**0,023**
Peak internal hip abduction moment (Nm/kg)	−0,79 ± 0,45	−0,75 ± 0,44	−0,78 ± 0,47	0,506	0,261	0,124
Peak resultant GRF (N/kg)	25,3 ± 6,7	24,4 ± 6,7	25,0 ± 7,2	0,073	0,721	**0,048**
Peak loading rate of res. GRF (N/kg/s)	516,9 ± 198,9^{TF**}	589,1 ± 246,8^{BF**}	563,1 ± 208,8	**0,015**	0,738	0,347

Note: Bold parameter names indicate a significant ANOVA effect ($p < 0.05$). *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$ revealing significant Bonferroni-corrected post-hoc tests. p-Value adjusted for comparing a family of 3.

Table 3
Mean ± 1 standard deviation of analyzed parameters for the running movement.

Parameter	Barefoot (BF)	Tape fresh (TF)	Tape used (TU)	p-Value Tape	p-Value Sex	p-Value Interaction
Peak ankle dorsiflexion angle (°)	−17,1 ± 5,1^{TF*}	−13,4 ± 5,8^{TU*}	−16,8 ± 6,6$^{BF*, TF*}$	**0,008**	0,172	0,731
Duration of rearfoot eversion (% stance)	62,5 ± 16,9	63,4 ± 18,9	62,2 ± 17,8	0,913	0,247	0,958
Peak ankle eversion angle (°)	−7,2 ± 4,2$^{TF*, TU**}$	−5,6 ± 3,5^{BF*}	−5,6 ± 3,9^{BF**}	**0,003**	0,743	0,282
Ankle inversion angle at touchdown (°)	5,4 ± 4,3	4,1 ± 3,8	5,3 ± 3,6	0,07	0,522	0,75
Ankle eversion range of motion (°)	−12,6 ± 4,9$^{TF***, TU*}$	−9,7 ± 3,0^{BF***}	−10,9 ± 3,5^{BF*}	**<0.001**	0,758	0,489
Knee flexion angle at touchdown (°)	22,2 ± 7,4	22,3 ± 8,8	21,9 ± 8,2	0,715	0,234	0,679
Peak knee adduction angle (°)	6,6 ± 4,2	6,3 ± 4,0	6,1 ± 4,5	0,605	0,654	0,999
Peak knee internal rotation angle (°)	15,0 ± 5,2	13,9 ± 5,0	13,8 ± 5,1	0,22	0,399	0,751
Peak internal knee abduction moment (Nm/kg)	−0,83 ± 0,34	−0,77 ± 0,38	−0,78 ± 0,38	0,243	0,208	0,528
Internal knee abduction angular impulse (Nm s/kg)	−0,08 ± 0,04	−0,07 ± 0,04	−0,07 ± 0,04	0,342	0,205	0,239
Average hip abduction moment (Nm/kg)	−0,94 ± 0,20^{TF*}	−0,90 ± 0,21^{BF*}	−0,93 ± 0,18	**0,037**	0,594	0,227
Peak hip adduction angle (°)	12,2 ± 5,7	11,4 ± 5,7	12,1 ± 5,9	**0,038**	0,186	0,376
Peak hip abduction moment (Nm/kg)	−2,02 ± 0,39	−1,97 ± 0,40	−2,04 ± 0,38	0,273	0,878	0,127
Peak hip internal rotation angle (°)	5,1 ± 5,4	4,6 ± 4,9	4,2 ± 4,6	0,437	0,177	0,455
Contralateral pelvic drop angle (°)	−4,7 ± 3,1	−4,2 ± 3,2	−4,8 ± 3,1	0,259	0,389	0,093
Contact time (s)	0,20 ± 0,02	0,20 ± 0,02	0,20 ± 0,02	0,145	0,057	0,101
Braking GRF impulse (m/s)	−0,24 ± 0,06	−0,23 ± 0,06	−0,25 ± 0,05	0,057	0,758	0,473
Peak resultant GRF (N/kg)	26,1 ± 3,2	26,1 ± 3,4	26,2 ± 3,3	0,713	0,963	0,524
Peak loading rate of res. GRF (N/kg/s)	726,8 ± 138,7$^{TF*, TU*}$	758,8 ± 167,4^{BF*}	763,7 ± 137,5^{BF*}	**0,015**	**<0.001**	0,096

Note: Bold parameter names indicate a significant ANOVA effect ($p < 0.05$). *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$ revealing significant Bonferroni-corrected post-hoc tests. p-Value adjusted for comparing a family of 3.

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3. Results

3.1. Change of direction

We found no interaction effects between tape and sex for any of the observed outcome parameters in COD; therefore, the reported statistical results for COD are for the main effect of tape and the between-subject factor sex only (Table 1).

At the ankle joint, we found a main effect of tape for the peak ankle dorsiflexion angle. For TF, a significantly reduced peak ankle dorsiflexion angle compared to BF ($d = 0.66$) was found (Table 1). No significant sex differences were found ($p > 0.05$). A main effect of tape was found for the peak ankle inversion angle. For both TF and TU, significantly reduced values to BF ($d = 1.02$ and $d = 0.76$, respectively) were found (Table 1). Women had significantly higher peak ankle inversion angles than men ($p < 0.05$, $d = 0.63$–1.02; Table S1). A main effect of tape for the peak internal ankle inversion moment was found (Table 1). For TU, a significantly reduced peak internal inversion moment compared to BF with an ES of $d = 0.44$ was found. No significant sex differences were found ($p > 0.05$; Table S1).

At the knee joint, we found no main effects of tape for the peak knee abduction angle and peak internal knee adduction moment. However, women had significantly higher peak knee abduction angles than men ($p < 0.05$, $d = 0.88$–1.38; Table S1). No significant sex differences for the peak internal knee adduction moment were found ($p > 0.05$; Table 1). We found no main effects of tape and no significant sex differences for peak resultant GRF (Table 1). A main effect of tape for the peak loading rate of the resultant GRF was found. For both TF and TU, significantly increased values to BF were found ($d = 0.45$ and $d = 0.56$ respectively). No significant sex differences were found ($p > 0.05$; Table S1).

At the hip joint, no statistically significant effects were identified (Table 1).

3.2. Drop jump

At the ankle joint, we found no main effect of tape and no sex differences for the peak ankle dorsiflexion angle ($p > 0.05$, Table 2). A main effect of tape was found for the peak ankle inversion angle (Table 2). For both TF and TU, significantly reduced values to BF were found ($d = 0.91$ and $d = 0.49$ respectively). Women had significantly lower peak ankle inversion angles than men ($p < 0.05$, $d = 0.17$–0.75; Tables 2, S2). A main effect of tape for the peak internal ankle inversion moment was found. For both TF and TU, significantly reduced values to BF were found with ES of $d = 0.36$ and $d = 0.41$, respectively. No significant sex differences were found ($p > 0.05$; Tables 2, S2).

At the knee joint, we found no main effect for the peak internal knee adduction moment (Table 2). However, women had significantly higher peak internal knee adduction moments than men ($p < 0.05$, $d = 0.74$–1.18; Table S2). We found tape * sex interaction effects for the peak knee abduction angle and peak resultant GRF in DJ ($p < 0.05$; Tables 2, S2). For the peak knee abduction angle, the interaction effect was caused by an increase in this parameter with TF and TU ($d = 0.29$ and $d = 0.54$, respectively) in females, while in males, these parameters remained similar ($d = 0.12$ and $d = 0.02$, respectively). For peak resultant GRF, the interaction effect was caused by a significant reduction in TF ($d = 0.35$) in females, while in males, no differences between tape conditions were found ($d < 0.05$). A main effect of tape for the peak loading rate of the resultant GRF was found ($p = 0.005$, Table 2). For TF, a significantly increased peak loading rate of the resultant GRF to BF was found ($d = 0.32$). No significant sex differences were found ($p > 0.05$; Table S2).

At the hip joint, we identified a significant tape-by-sex interaction effect ($p = 0.023$, Table 2) for peak hip adduction angles. This effect was caused by women changing their hip adduction more toward abduction, whereas men changed more toward adduction during the tape conditions (Table S1). Peak internal hip abduction moments were not significantly affected by any of the investigated factors (Table 2).

3.3. Running

At the ankle joint, we found a main effect of tape for the peak ankle eversion angle (Table 3). For both TF and TU, significantly reduced values to BF ($d = 0.42$ and $d = 0.39$, respectively) were found. No significant sex differences were found ($p > 0.05$; Table S3). A main effect of tape for ankle eversion ROM was found (Table 3). For both TF and TU, significantly reduced values to BF ($d = 0.73$ and $d = 0.42$, respectively) were found. No significant sex differences were found ($p > 0.05$; Table S3). We found no main effects or sex differences for the duration of rearfoot eversion and ankle inversion at touchdown ($p > 0.05$; Tables 3, S3).

At the knee joint, we found no main effects or sex differences for any of the observed outcome parameters ($p > 0.05$; Table 3).

At the hip joint, we found a main effect of tape for the peak hip adduction angle ($p = 0.038$, Table 3). For TF, significantly reduced values to BF and TU ($d = 0.15$ and $d = 0.13$, respectively) were found. No significant sex differences were found ($p > 0.05$; Tables 3, S3). No main effects of tape or sex differences were found for peak hip internal rotation and contralateral pelvic drop angles (Table 3). We found a main effect for the average hip abduction moment (Table 3). For TF, a significantly reduced average hip abduction moment to BF ($d = 0.22$) was found. No significant sex differences were found ($p > 0.05$; Tables 3, S3). We found no main effect of tape or sex differences for the peak hip abduction moment ($p > 0.05$; Tables 3, S3).

We found a tape-by-sex interaction effect for the peak loading rate of the resultant GRF ($p < 0.05$; Tables 3, S3). This interaction effect was caused by a more pronounced increase in TF in females ($d = 0.53$) compared to male participants ($d = 0.07$). Finally, we found no main effects of tape or sex differences for peak resultant GRF, braking GRF impulse, and contact time ($p > 0.05$; Tables 3, S3).

The descriptive statistics for all analyzed parameters and movements, separated by sex, can be found in Tables S1 to S3 of the Supplementary digital content. Figures of the average waveform patterns for the analyzed joint angles and moments can be found in Figs. S1 to S6 of the Supplementary digital content.

4. Discussion

The purpose of this study was to determine whether there are changes in the kinetics and kinematics related to traumatic or overuse lower extremity joint injuries during sports-specific tasks with and without taping during barefoot movements and further, to determine whether there is a change in the taping-induced kinetic and kinematic alterations of the lower extremities obtained after using the tape for a specific time.

The results show that ankle taping significantly restricted the peak ankle inversion for all dynamic movements (DJ TF: −3.2°; COD TF: −4.1°; DJ TU: −1.9°; COD TU: −2.8°) and the peak ankle dorsiflexion for COD and RUN movements (COD TF: −4.1°; RUN TF: −3.7°; COD TU: −2.0°; RUN TU: −0.3°). While not significant, a reduction in peak ankle dorsiflexion was also observed for DJ (TU: −2.3°). This finding is relevant since it, in theory, might explain the increased maximum loading rate of the GRF observed with taping in all analyzed movements. Ankle taping is generally considered to restrict the passive range of motion at the ankle, even though this effect reduces over time (for review, see 24). Fewer studies have assessed taping effects during sports-specific movements. Some found reduced peak ankle inversion angles,[25] while others did not identify such effects.[26] However, differences in the movements analyzed and taping techniques used, the use of footwear, and the methodological procedures used to quantify ankle joint motion can affect the experimental outcome, which might explain the inconclusive evidence across studies.

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Reducing ankle motion in the frontal and sagittal planes reduces the ability to absorb impact energy during eccentric work of muscle–tendon-units spanning the ankle, thus creating a stiffer ankle joint. Based on fundamental mechanical theory, increasing the stiffness in a mechanical system leads to higher loading rates when subjected to the same loading impulse. Restricted ankle motion during sports-specific movements has been identified before, e.g., during drop jump landings[27] or change of direction tasks.[28] We believe that increasing loading rates might expose athletes to an increased risk of overuse injuries. This is because peak loading rates have been identified to be a relevant risk factor for plantar fasciitis in distance runners.[7] The findings regarding the ankle joint led us to accept our first hypothesis that ankle joint excursions would be significantly reduced with freshly applied tape.

We further hypothesized that the restriction effects of taping on ankle ROM would alter the kinetic parameters associated with traumatic or overuse injury risk. Taping at the ankle joint resulted in lower ankle frontal plane peak joint moments compared to barefoot (COD TF: −0.08 Nm/kg; COD TU: −0.11 Nm/kg; DJ TF: −0.14 Nm/kg; DJ TU: −0.15 Nm/kg), which aligns with previous research.[25,30] At the knee joint, no significant changes were induced by taping in peak internal knee adduction moments, previously associated with ACL injury risk in team sports.[31,32] Furthermore, taping did not induce significant changes in peak internal knee abduction moments and impulses during RUN. These parameters have been associated with the risk of suffering from patellofemoral pain syndrome in distance runners.[7] Furthermore, the knee flexion angle at touchdown, which has also been associated with ACL injury risk[33] and the risk of sustaining iliotibial band syndrome in distance running,[7] was not significantly different in TF or TU during RUN. However, we identified a significant reduction in the average hip abduction moment, a parameter associated with the risk of sustaining patellofemoral pain syndrome in distance running, for TF during RUN. Ihmels et al. also reported no significant effect of taping on knee moments in their study.[34] In addition, Williams et al. showed that the peak knee flexion moment decreased during the cutting task, and internal rotation moment and medial shear force decreased during a rebound task, but the peak knee internal rotation moment increased during a cutting task.[35] Stoffel et al. highlighted that the taped condition tended to increase the peak valgus moment and decrease the peak internal knee moment and peak varus moment during running and sidestep tasks compared to no taping.[25] It is also worth noting that in our study, the tape did not change the peak knee extension moment during the RUN movement at baseline, whereas it increased this moment with the use of the tape.

As in previous research, reducing peak ankle joint angle excursions in the sagittal and frontal plane using tape increased GRF loading rates.[36,37] Increased loading rates can be attributed to an increased risk of overuse injuries.[7] For RUN, we further analyzed parameters that have been associated with overuse injury risk.[7] During this analysis, in addition to the parameters described above, we found significant reductions in peak ankle eversion angles and eversion range of motion during RUN, two parameters that have been related to medial tibial stress syndrome, Achilles tendinopathy, and tibial stress fractures.[7] Reduced ankle eversion angles during running with taping have been found in a previous study to a similar extent (−4.8°).[39]

Overall, while the increases in GRF loading rates across movements might be related to an increased overuse injury risk, particularly for plantar fasciitis, most biomechanical risk factors were reduced or unchanged when moving with taped ankles. These results led us to only partially accept our second hypothesis that taping would affect lower extremity joint kinetics.

Regarding our third hypothesis, that taping effects would be reduced after 15 min of sports-specific use, we observed that the restrictive effect of ankle taping on the peak ankle inversion angle during all movements mainly persisted despite using tape after our sports-specific motion protocol. Peak ankle inversion increased by about 1.2° from TF to TU ($d = 0.32$ and $d = 0.40$ for COD and DJ, respectively). Whether this increase is practically relevant for ankle sprain injury risk cannot be evaluated given the current status of the literature. Nonetheless, future studies should explore the time course of reducing the protective effects of ankle taping with larger spatial distribution and for a longer time than 15 min. Furthermore, the magnitudes of meaningful changes in peak ankle inversion should be targeted for an improved interpretation of findings in studies on protective technologies like taping or bracing. The significant restrictive effect of taping on peak ankle dorsiflexion during COD and RUN disappeared more clearly with the use of tape. However, peak resultant GRF loading rates remained increased in TU compared to BF in COD and RUN. This finding challenges the assumption that restricting peak dorsiflexion is a key mechanism underlying the increased loading rates when wearing tape. Future studies need to elaborate on the relationship between stiffening the ankle joint in different planes of motion and the effects of such stiffening interventions on GRF loading rates in greater detail. After 15 min of sports-specific use, the alterations induced by TF remained for some biomechanical risk factors while they became closer to BF again for others, indicating a differential response to prolonged use of taping for different biomechanical variables. Such a differential response might be the effect of differential stresses experienced by the individual components of the tape. The stirrups would probably be the last to be affected, as many layers of interlocking strips secure their integrity. Since the stirrups' function is to restrict the ankle range of motion in the frontal plane, it seems conceivable that the restrictive effect persists longer for the frontal than the sagittal plane when sports-specific movements stress the tape. We did not analyze the material integrity of the tapes after their usage in the current study. Therefore, this interpretation remains speculative at the moment. However, future studies might include such analyses to create a better understanding of the relationship between taping technologies and protection effects after prolonged use.

Despite several strengths, this study also has some limitations. Although several sports-specific movements can cause ankle sprains, we studied only three movements. This is because the measurement time increases as the number of tasks increases, and thus, fatigue might pose a critical factor. Another limitation is that we used only one taping technique for ankle support. However, this choice was motivated by the frequent use of this taping technique in research studies and sports practice. Adding other taping techniques or supports would have enriched the study but would have required more data collection and shifted the focus of the study away from it. Furthermore, the study was conducted in healthy volunteers with no lower limb complaints. A study in athletes with chronic ankle instability might have presented different effects on lower limb biomechanics. We also analyzed the participants barefoot to isolate the taping effects on lower extremity biomechanics. Nonetheless, the actual sport is performed shod. Therefore, future studies should explore how these isolated taping effects are altered when wearing different types of athletic footwear. Finally, the tape in this study was applied to protect the ankle joint from excessive joint excursions. However, in a professional setting, the tape might be applied "submaximally" to keep the players' performance at a relatively high level by allowing for more range of motion to perform performance-relevant tasks better than with maximally restricted ankles. To provide a more holistic understanding of ankle protection technologies, it is therefore recommended to include performance and subjective perception parameters in an analysis of taping techniques in the future.[22]

5. Conclusion

This study aimed to investigate the impact of ankle taping on lower extremity joint kinetics and kinematics during sports-specific movements and how these effects change after sports-specific use of the tape. Our findings suggest that ankle taping effectively reduces peak ankle inversion angles and thereby provides protection against ankle sprains during

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dynamic activities. Furthermore, most biomechanical risk factors for ACL or running overuse injuries either remained unchanged or decreased with fresh tape application, with the exception of the peak loading rate of the resultant ground reaction force, which increased between 4 % and 18 % between movements. The results indicate that the immediate restrictive effects of ankle taping are beneficial in reducing peak ankle inversion and dorsiflexion angles, potentially mitigating the risk of ankle sprains. However, the increased peak loading rate of the resultant ground reaction force across all movements suggests a possible elevated risk for overuse injuries such as plantar fasciitis. This highlights the need for further investigation into the long-term implications of increased loading rates due to ankle taping. The protective effects of ankle taping were observed to diminish after 15 min of sports-specific use, particularly for peak ankle dorsiflexion during change of direction and running movements. However, the restrictive effects on peak ankle inversion angles diminished to a lesser extent. This differential response underscores the necessity for future research to explore the duration and consistency of taping's protective effects, as well as its influence on different biomechanical variables over extended periods and across various sports.

In conclusion, ankle taping offers significant protection against ankle sprains without substantially increasing the risk of ACL or running overuse injuries. The findings support the use of ankle taping as a preventive measure in sports, although the potential increase in loading rates warrants caution and further study. Future research should focus on the long-term effects of ankle taping, its impact on athletes with chronic ankle instability, and the interaction between taping and different types of athletic footwear to develop a comprehensive understanding of its benefits and limitations.

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Confirmation of ethical compliance

The study procedures were approved by the ethics committee of Offenburg University of Applied Sciences, Offenburg, Germany. The Declaration of Helsinki was strictly followed throughout the study.

CRediT authorship contribution statement

The study was designed and planned by B.U., S.W. and P.M. Participant recruitment was performed by G.B., J.H., H.K. and Y.D. Experimental data were collected by B.U., J.H., G.B., H.K., M.H. and L.B. Postprocessing of collected data was performed by M.H., G.B., H.K., F.P., J.H. and B.U. Model calculations were performed by S.W. and B.U. Figures and tables were created by B.U., L.B. and F.P. The paper was written by B.U. with guidance by S.W. All authors reviewed, corrected and approved the initial version of the manuscript.

Declaration of interest statement

The authors state that they have no competing interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jsams.2024.07.002.

References

1. Megalaa T, Hiller CE, Ferreira GE et al. The effect of ankle supports on lower limb biomechanics during functional tasks: a systematic review with meta-analysis. *J Sci Med Sport* 2022;25(7):615-630. doi:10.1016/j.jsams.2022.02.008.
2. McKay GD. Ankle injuries in basketball: injury rate and risk factors. *Br J Sports Med* 2001;35(2):103-108. doi:10.1136/bjsm.35.2.103.
3. Doherty C, Delahunt E, Caulfield B et al. The incidence and prevalence of ankle sprain injury: a systematic review and meta-analysis of prospective epidemiological studies. *Sports Med* 2014;44(1):123-140. doi:10.1007/s40279-013-0102-5.
4. Dizon JMR, Reyes JJB. A systematic review on the effectiveness of external ankle supports in the prevention of inversion ankle sprains among elite and recreational players. *J Sci Med Sport* 2010;13(3):309-317. doi:10.1016/j.jsams.2009.05.002.
5. De Ridder R, Willems TM, Vanrenterghem J et al. Effect of tape on dynamic postural stability in subjects with chronic ankle instability. *Int J Sports Med* 2015;36(4):321-326. doi:10.1055/s-0034-1385884.
6. Chinn L, Dicharry J, Hart JM et al. Gait kinematics after taping in participants with chronic ankle instability. *J Athl Train* 2014;49(3):322-330. doi:10.4085/1062-6050-49.3.08.
7. Willwacher S, Kurz M, Robbin J et al. Running-related biomechanical risk factors for overuse injuries in distance runners: a systematic review considering injury specificity and the potentials for future research. *Sports Med* 2022;52(8):1863-1877. doi:10.1007/s40279-022-01666-3.
8. Lysdal FG, Wang Y, Delahunt E et al. What have we learnt from quantitative case reports of acute lateral ankle sprains injuries and episodes of 'giving-way' of the ankle joint, and what shall we further investigate? *Sports Biomech* February 21, 2022:1-21. doi:10.1080/14763141.2022.2035801. [published online].
9. Hewett TE, Myer GD, Ford KR et al. Biomechanical measures of neuromuscular control and valgus loading of the knee predict anterior cruciate ligament injury risk in female athletes: a prospective study. *Am J Sports Med* 2005;33(4):492-501. doi:10.1177/0363546504269591.
10. Hewett T, Myer G. The mechanistic connection between the trunk, hip, knee, and anterior cruciate ligament injury. *Exerc Sport Sci Rev* 2011;39(4):161-166. doi:10.1097/JES.0b013e3182297439.
11. Wilkerson GB. Comparative biomechanical effects of the standard method of ankle taping and a taping method designed to enhance subtalar stability. *Am J Sports Med* 1991;19(6):588-595. doi:10.1177/036354659101900606.
12. Forbes H, Thrussell S, Haycock N et al. The effect of prophylactic ankle support during simulated soccer activity. *J Sport Rehabil* 2013;22(3):170-176. doi:10.1123/jsr.22.3.170.
13. Cordova ML, Ingersoll CD, Palmieri RM. Efficacy of prophylactic ankle support: an experimental perspective. *J Athl Train* 2002;37(4):446-457.
14. Eils E, Imberge S, Völker K et al. Passive stability characteristics of ankle braces and tape in simulated barefoot and shoe conditions. *Am J Sports Med* 2007;35(2):282-287. doi:10.1177/0363546506294471.
15. Jewell C, Trudeau MB, Rohr E et al. Calcaneal movement measured by skin versus shoe-mounted markers. *Footwear Sci* 2017;9(sup1):S72-S73. doi:10.1080/19424280.2017.1314345.
16. Reinschmidt C, van den Bogert A, Lundberg A et al. Tibiofemoral and tibiocalcaneal motion during walking: external vs. skeletal markers. *Gait Posture* 1997;6:98-109. doi:10.1016/S0966-6362(97)01110-7.
17. Abián-Vicén J, Alegre LM, Fernández-Rodríguez JM et al. Ankle taping does not impair performance in jump or balance tests. *J Sports Sci Med* 2008;7(3):350-356.
18. Willwacher S, Kurz M, Menne C et al. Biomechanical response to altered footwear longitudinal bending stiffness in the early acceleration phase of sprinting. *Footwear Sci* 2016;8(2):99-108. doi:10.1080/19424280.2016.1144653.
19. Söderkvist I, Wedin PÅ. Determining the movements of the skeleton using well-configured markers. *J Biomech* 1993;26(12):1473-1477. doi:10.1016/0021-9290(93)90098-Y.
20. Mai P, Willwacher S. Effects of low-pass filter combinations on lower extremity joint moments in distance running. *J Biomech* 2019;95:109311. doi:10.1016/j.jbiomech.2019.08.005.
21. Kristianslund E, Krosshaug T, van den Bogert AJ. Effect of low pass filtering on joint moments from inverse dynamics: implications for injury prevention. *J Biomech* 2012;45(4):666-671. doi:10.1016/j.jbiomech.2011.12.011.
22. Willwacher S, Bruder A, Robbin J et al. A multidimensional assessment of a novel adaptive versus traditional passive ankle sprain protection systems. *Am J Sports Med* 2023;51(3):715-722. doi:10.1177/03635465221146294.
23. Cohen J. *Statistical Power Analysis for the Behavioral Sciences2nd ed.*, 1988. doi:10.4324/9780203771587. Routledge.
24. Verhagen EALM, van der Beek AJ, van Mechelen W. The effect of tape, braces and shoes on ankle range of motion. *Sports Med* 2001;31(9):667-677. doi:10.2165/00007256-200131090-00003.
25. Stoffel KK, Nicholls RL, Winata AR et al. Effect of ankle taping on knee and ankle joint biomechanics in sporting tasks. *Med Sci Sports Exerc* 2010;42(11):2089-2097.
26. Koya M, Ishiguro K, Nakayama T. Immediate effects of two different ankle taping techniques on ankle inversion during quick lateral cutting movements; a pilot cross-over randomised controlled trial. *J Phys Ther Sports Med* 2018;2(1):5-9. [6 J Phys Ther Sports Med 2018 2 Issue. 2018;1:3].
27. Cordova ML, Takahashi Y, Kress GM et al. Influence of external ankle support on lower extremity joint mechanics during drop landings. *J Sport Rehabil* 2010;19(2):136-148. doi:10.1123/jsr.19.2.136.
28. Meana M, Alegre LM, Elvira JLL et al. Kinematics of ankle taping after a training session. *Int J Sports Med* 2008;29(01):70-76. doi:10.1055/s-2007-965126.
29. Sato N, Nunome H, Hattori M et al. Effect of individual ankle taping components on the restriction of ankle external inversion moment. *Sports Biomech* 2023;22(9):1181-1191. doi:10.1080/14763141.2020.1792539.

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30. Sigurðsson HB, Karlsson J, Snyder-Mackler L et al. Kinematics observed during ACL injury are associated with large early peak knee abduction moments during a change of direction task in healthy adolescents. *J Orthop Res* 2021;39(10):2281-2290. doi:10.1002/jor.24942.
31. Myer GD, Ford KR, Stasi SLD et al. High knee abduction moments are common risk factors for patellofemoral pain (PFP) and anterior cruciate ligament (ACL) injury in girls: is PFP itself a predictor for subsequent ACL injury? *Br J Sports Med* 2015;49(2):118-122. doi:10.1136/bjsports-2013-092536.
32. Yu B, Garrett WE. Mechanisms of non-contact ACL injuries. *Br J Sports Med* 2007;41(Suppl 1):i47-i51. doi:10.1136/bjsm.2007.037192.
33. Ihmels WD, Seymore KD, Brown TN. Effect of sex and ankle brace design on knee biomechanics during a single-leg cut. *Am J Sports Med* 2020;48(6):1496-1504. doi:10.1177/0363546520911048.
34. Williams SA, Ng L, Stephens N et al. Effect of prophylactic ankle taping on ankle and knee biomechanics during basketball-specific tasks in females. *Phys Ther Sport* 2018;32:200-206. doi:10.1016/j.ptsp.2018.04.006.
35. de CN SI, Takahasi HY, Suda EY et al. Ground reaction force in basketball cutting maneuvers with and without ankle bracing and taping. *Sao Paulo Med J* 2006;124(5):245-252. doi:10.1590/S1516-31802006000500002.
36. Hodgson B, Tis L, Cobb S et al. The effect of external ankle support on vertical ground-reaction force and lower body kinematics. *J Sport Rehabil* 2005;14(4):301-312. doi:10.1123/jsr.14.4.301.
37. De Clercq DLR. Ankle bracing in running: the effect of a Push® type medium ankle brace upon movements of the foot and ankle during the stance phase. *Int J Sports Med* 1997;18(3):222-228. doi:10.1055/s-2007-972624.

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